

Basic Trig Rules

section 2.2

9-2-25

Example:

find a and b so f is
differentiable EVERYWHERE.

$$f(x) = \begin{cases} ax^2 + 1, & x \leq 2 \\ bx - 3, & x > 2 \end{cases}$$

$$\lim_{x \rightarrow 2^-} ax^2 + 1 = \lim_{x \rightarrow 2^+} bx - 3 \quad (1)$$

$$4a + 1 = 2b - 3 \quad (2)$$

$$2ax = b \quad (3)$$

$$4a = b$$

$$4a + 1 = 2b - 3 \quad (4)$$

$$4a + 1 = 8a - 3$$

$$a = 1$$

$$b = 4$$

(5)

Sine and Cosine Rule

$$\frac{d}{dx} [\sin x] = \cos x$$

$$\frac{d}{dx} [\cos x] = -\sin x$$

Ex:

$$y = 6x^2 + 5 \cos x$$

$$y' = 12x - 5 \sin x \quad \checkmark$$

a cool tid-bit

$$y'' = \text{a double derivation}$$

$$y''' = \text{a triple derivation}$$

$$y^{(4)}$$

and so on...

$$\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} = f'(x)$$

It's not really a limit bro

$$f(x) = \sin x$$

So...

$$f'(x) = \cos x$$



coolio

Ex:

Table Problems

$$h(x) = 4(x) + 5g(x) \text{ when } x=2$$

find $h'(2)$

$$h'(x) = f'(x) + 5g'(x)$$

$$h'(2) = f'(2) + 5g'(2)$$

$$f'(2) = -1 \quad 5g'(2) = 5(0)$$

$$h'(x) = -1 + 5(0)$$

$$h'(x) = -1 \quad \checkmark$$

find $h'(x)$

$$h(x) = g(x) - \cos(x) \text{ @ } x=0$$

$$h'(x) = g'(x) + \sin(x)$$

$$g'(0) = 4 \quad \sin(0) = 0$$

$$h'(x) = 4 + 0$$

$$h'(x) = 4 \quad \checkmark$$